

Simplifying algebraic expressions



- 1 a -18 b -1 c $\frac{1}{3}$ d $\frac{6}{7}$
- 2 $9x^2$
- 3 a No b Yes c Yes d Yes
- 4 a Yes b No c No d No
- 5 a Yes b No c Yes d No
- 6 a $4n$ b $24ab$ c $-11x$ d $-11m$
e $-18n$ f $x + 4y$ g $2j + 2$ h $3w$
i $-7a$ j $-12q$ k $7m^2 + 9m$ l $-72atu$
m $5m$ n $2y - 11$ o $20x$ p $5x$
q $4 + 2x$ r $-6y + 33$ s $3x$ t $12xy + 16z + 1$
u $-4x + 5y + 17$ v $15 - p$ w $17p$
- 7 a $14x$ b $2y - 36$ c $5y + 32$ d $-10x + 81$
e $8y - 14$ f $16y - 1$ g $7y + 1$ h $-y + 49$
i $4x - 28$ j $72x - 10$ k $2y - 63$ l $7x + 57$
m $18x + 60$ n $-2y + 73$ o $-10a + 108$ p $15x - 54$
q $-\frac{8x}{3} - 77$ r $-8x + y$ s $14x - 22y - 2$
- 8 $4x + 2$
- 9 a $22x + 36$ m b $110x + 180$ m



10 19×19

11 $x^2 \times x^2 \times x^2 \times x^2 \times x^2 \times x^2$

12

a $u + u + u + b$ dollars

b $3u + b$

13

a $x + y$

b $6x + 5y$

14

a Yes

b Yes

c No

d No

e No

f Yes

15 Possible answers are:

$b + 0.10b$

$1.1b$

16

a Step (3)

b $10xy + 2x + y + 10$

17

a Step (1)

b $2x^2 + 5y + 9x^2y$

Additional questions

18

a $9u$

b $4a$

c 3

d 4

e $-35n$

f $\frac{3w}{8}$

g $21v$

h $6 + 5r$

i $-15v$

j $14v$

k $-13t$

l $\frac{16t}{27}$

m $8m^2$

n $-43n$

o 2

p $30t$

a $12m$

r $13v$

s $15m + 10$

t 3

$$u \quad 6$$

$$y \quad 18n^2$$

$$v \quad 2$$

$$z \quad 3$$



$$9n^2$$

$$x \quad -m$$

19

$$a \quad 34t$$

$$e \quad 8n^2$$

$$i \quad 12w$$

$$m \quad 8p^2$$

$$q \quad \frac{3x}{2}$$

$$u \quad \frac{3x+2}{2}$$

$$y \quad \frac{5x+2}{12}$$

$$b \quad 27c$$

$$f \quad 27n^2$$

$$j \quad 24pq$$

$$n \quad 10s$$

$$r \quad -\frac{8y}{11}$$

$$v \quad \frac{p}{4}$$

$$z \quad \frac{7x+9}{6}$$

$$c \quad 5a$$

$$g \quad 22n^2$$

$$k \quad 4n$$

$$o \quad 2u$$

$$s \quad \frac{31y}{20}$$

$$w \quad \frac{17p}{35}$$

$$d \quad 36a$$

$$h \quad 13mn^2$$

$$l \quad 12k^2$$

$$p \quad 6k^2$$

$$t \quad \frac{2x}{3}$$

$$x \quad \frac{11p}{15}$$

20

$$a \quad 9$$

$$b \quad \frac{4w}{9}$$

21

$$a \quad 12$$

$$b \quad \frac{m}{6}$$

22

$$a \quad (4 + 16n) \text{ cm}$$

$$e \quad 24v \text{ cm}$$

$$i \quad (23d - 12) \text{ units}$$

$$b \quad (14 + 2q) \text{ m}$$

$$f \quad 8a \text{ cm}$$

$$c \quad 15b \text{ m}$$

$$g \quad (22n - 1) \text{ units}$$

$$d \quad 24v \text{ cm}$$

$$h \quad (27x + 10) \text{ units}$$

23

$$a \quad 42v \text{ mm}^2$$

$$e \quad 36q^2 \text{ m}^2$$

$$b \quad 6q \text{ m}^2$$

$$f \quad 16n^2 \text{ m}^2$$

$$c \quad 50q^2 \text{ cm}^2$$

$$d \quad 25n^2 \text{ cm}^2$$

24

$$a \quad 56a^2 \text{ mm}^3$$

$$b \quad (16a^2 + 84a) \text{ mm}^2$$

$$25 \quad \frac{2x + y + z}{2}$$

26

a

$$\frac{3x - y}{9} \text{ dollars}$$



b

$$\frac{3x - y + 11}{9} \text{ dollars}$$

$$27 \quad 120x^2 \text{ cm}^2$$